

test_atomic_gen_constrained.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/atm/test_atomic_gen_constrained.vr

```
//
// Test the atomic generator
//
#ifndef SYNOPSIS_NTB
#include "atm.vrh"
#endif

class atm_cell_with_better_vpi extends atm_cell {
    constraint better_vpi {
        vpi dist {0:/1, 1:254:/1, 255:/1};
    }
}

class some_bad_atm_cell extends atm_cell {
    constraint hec_valid { // Some bad cells
        hec dist {0:/1, 1:255:/1};
    }
}

program test_gen {
    atm_cell_atomic_gen gen = new("Singleton");
    atm_cell cell;
    atm_cover cvr = new;
    fork
    {
        while (1) {
            cell = gen.out_chan.get_t();
            cell.display();
            cvr.cover(cell);
        }
    }
    join none

    {
        atm_cell_with_better_vpi my_cell = new;
        gen.randomized_obj = my_cell;
    }

    rvm_note(gen.log, "Generating 10 cells...");
    gen.stop_after_n_insts = 10;
    gen.start_xactor();
}
```

```
void = gen.notify.wait_for_t(gen.DONE);
delay(100);
gen.reset_xactor();
// Replace the constraints with another set
{
    some_bad_atm_cell my_cell = new;
    gen.randomized_obj = my_cell;
}

rvm_note(gen.log, "Generating 10 more cells...");
gen.stop_after_n_insts = 10;
gen.start_xactor();
void = gen.notify.wait_for_t(gen.DONE);
delay(100);
gen.log.report();
}
```